

Value of Information Analysis

METIMMOX-1 Economic Evaluation

Ben Geisler

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1 Overview

This report presents the results of an Expected Value of Partially Perfect Information (EVPPI) analysis for the METIMMOX-1 decision-analytic model. The analysis evaluates the value of reducing uncertainty in individual model parameters across multiple scenarios with varying willingness-to-pay (WTP) thresholds and nivolumab drug costs.

Key Feature: This report compares EVPI and EVPPI results across three different model structures (Model A, B, C) to assess the robustness of value of information findings to modeling assumptions about how biomarker interactions are specified.

The Expected Value of Partially Perfect Information (EVPPI) quantifies the value of eliminating uncertainty in specific model parameters. It represents the maximum amount that should be paid for additional research to perfectly inform a particular parameter. EVPI (Expected Value of Perfect Information) represents the value of eliminating all uncertainty, and EVPPI for any parameter cannot exceed total EVPI.

2 Model Structure Definitions

We compare three model structures to assess the robustness of value of information results to the specification of biomarker-treatment interactions in the survival models.

Table 1: Model structure definitions used in the analysis

Model	Name	Formula Pattern	Description
Model A	Joint	Age + sex + Rx + crp:Rx + tlr:Rx + tmb_braf:Rx	All biomarkers and all treatment interactions in one unified model
Model B	Focused	Age + sex + Rx + crp + tmb_braf + crp:Rx + tmb_braf:Rx	Per-strategy formulas: CRP/TMB_BRAF with both interactions, TLR with TLR-only interaction
Model C	Separate	Age + sex + Rx + [biomarker]:Rx	Only one biomarker and its treatment interaction per model (simplest specification)

Rationale: Model A (Joint) is the most comprehensive specification but may suffer from overfitting with limited sample size. Model C (Separate) is the simplest but ignores potential confounding from other biomarkers. Model B (Focused) uses a shared formula with CRP and TMB/BRAF (excluding TLR), providing simultaneous estimates for both interactions while adjusting for each other. Comparing results across all three structures provides insight into the structural uncertainty of our findings.

3 Scenario Definitions

Table 2: Analysis scenario definitions. Higher WTP thresholds (EUR100,000 and EUR150,000) are arbitrary exploratory thresholds, though the highest severity adjustment in Norwegian guidelines may fall between these values. The decreased nivolumab cost (EUR4,641) represents an 80% discount compared to the 40% discount in the base case scenario.

Scenario Name	WTP Threshold	Nivolumab Cost
Base Case	EUR 51,000	EUR13,923
Decreased Nivolumab Cost	EUR 51,000	EUR 4,641
WTP €100,000	EUR100,000	EUR13,923
WTP €100,000 + Decreased Nivolumab	EUR100,000	EUR 4,641
WTP €150,000	EUR150,000	EUR13,923
WTP €150,000 + Decreased Nivolumab	EUR150,000	EUR 4,641

4 Base Case Scenario

Willingness-to-Pay Threshold: EUR51,000

Nivolumab Cost: EUR13,923

4.1 EVPI Comparison Across Model Structures

Table 3: Expected Value of Perfect Information (EVPI) by model structure - Base Case

	Model	EVPI
5	Model A: Joint	EUR1339.56
56	Model B: Focused	EUR1438.25
57	Model C: Separate	EUR922.7

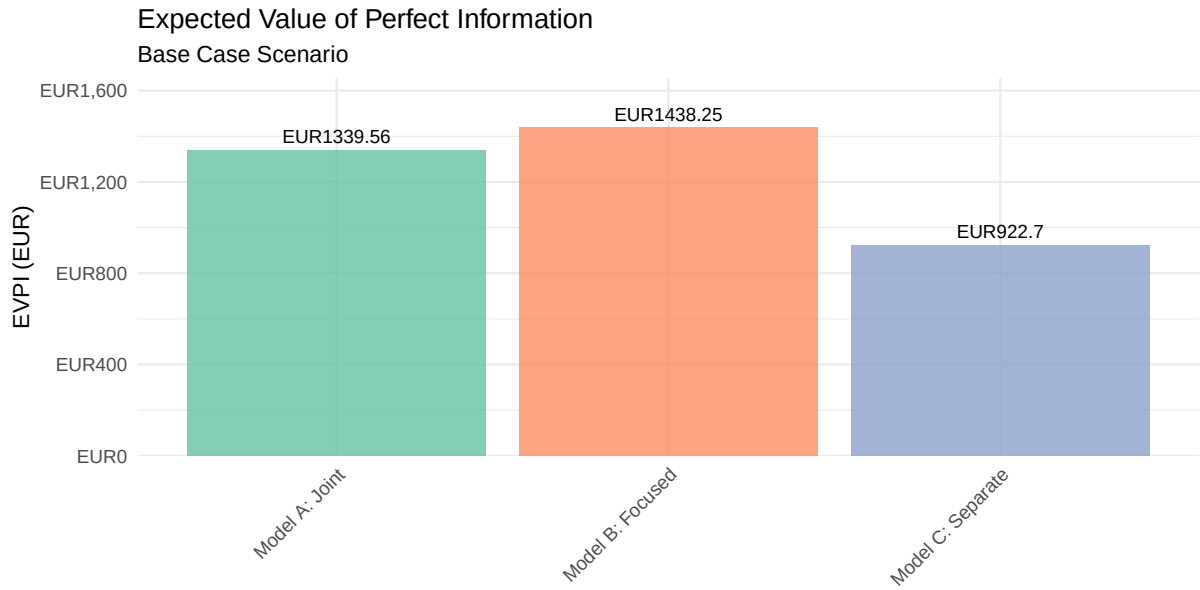


Figure 1: EVPI comparison across model structures (Base Case)

4.2 EVPPI Results by Model Structure

Table 4: EVPPI results by model structure - Base Case (ranked by average importance)

Parameter	EVPPi (A)	% (A)	EVPPi (B)	% (B)	EVPPi (C)	% (C)
u_np	EUR1064.09	79.4%	EUR941.75	65.5%	EUR888.69	96.3%
u_p	EUR889.04	66.4%	EUR818.14	56.9%	EUR779.14	84.4%
c_drug_FLOX	EUR194.26	14.5%	EUR1.27	0.1%	EUR61.48	6.7%
c_other_last	EUR87.67	6.5%	EUR53.07	3.7%	EUR108.87	11.8%
c_test_blood	EUR57.7	4.3%	EUR32.57	2.3%	EUR92.96	10.1%
c_other_visit	EUR0	0%	EUR119.68	8.3%	EUR0	0%
c_drug_nivo	EUR46.41	3.5%	EUR0	0%	EUR56.88	6.2%
c_other_baseline	EUR59.44	4.4%	EUR44.21	3.1%	EUR0	0%
c_test_CT	EUR12.04	0.9%	EUR0	0%	EUR37.88	4.1%
c_test_NGS	EUR0	0%	EUR0	0%	EUR0	0%
c_other_follow	EUR0	0%	EUR0	0%	EUR0	0%

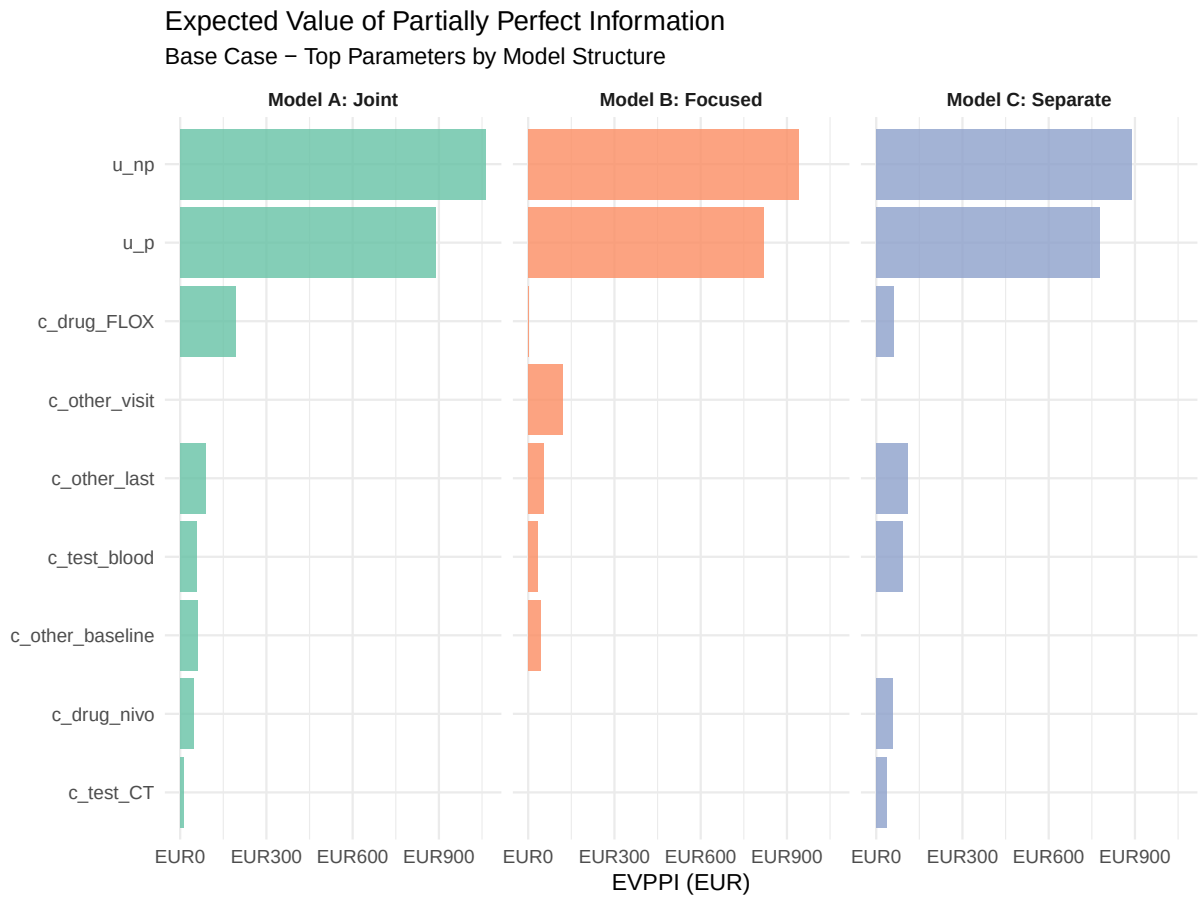


Figure 2: EVVPI by parameter and model structure (Base Case)

The positive EVVPI indicates decision uncertainty at the base case WTP threshold. This suggests that additional research to reduce parameter uncertainty could have value. The comparison across model structures shows whether this finding is robust to the specification of biomarker-treatment interactions.

5 Scenario 1: Decreased Nivolumab Cost

Willingness-to-Pay Threshold: EUR51,000

Nivolumab Cost: EUR4,641 (80% discount)

5.1 EVPI Comparison Across Model Structures

Table 5: Expected Value of Perfect Information (EVPI) by model structure - Scenario 1

	Model	EVPI
51	Model A: Joint	EUR8696.69
511	Model B: Focused	EUR8946
512	Model C: Separate	EUR8537.63

5.2 EVPPI Results by Model Structure

Table 6: EVPPI results by model structure - Scenario 1: Decreased Nivolumab Cost

Parameter	EVPPI (A)	% (A)	EVPPI (B)	% (B)	EVPPI (C)	% (C)
u_np	EUR877.52	10.1%	EUR890.43	10%	EUR947.52	11.1%
u_p	EUR646.44	7.4%	EUR746.64	8.3%	EUR762.97	8.9%
c_other_last	EUR111.93	1.3%	EUR91.82	1%	EUR68.49	0.8%
c_other_baseline	EUR15.85	0.2%	EUR121.8	1.4%	EUR112.98	1.3%
c_drug_FLOX	EUR0	0%	EUR95.01	1.1%	EUR48.47	0.6%
c_test_NGS	EUR56.33	0.6%	EUR30.21	0.3%	EUR1.68	0%
c_other_visit	EUR0	0%	EUR18.43	0.2%	EUR68.43	0.8%
c_drug_nivo	EUR0	0%	EUR56.71	0.6%	EUR0	0%
c_test_CT	EUR29.54	0.3%	EUR0	0%	EUR24.74	0.3%
c_test_blood	EUR0.71	0%	EUR0	0%	EUR0	0%
c_other_follow	EUR0	0%	EUR0	0%	EUR0	0%

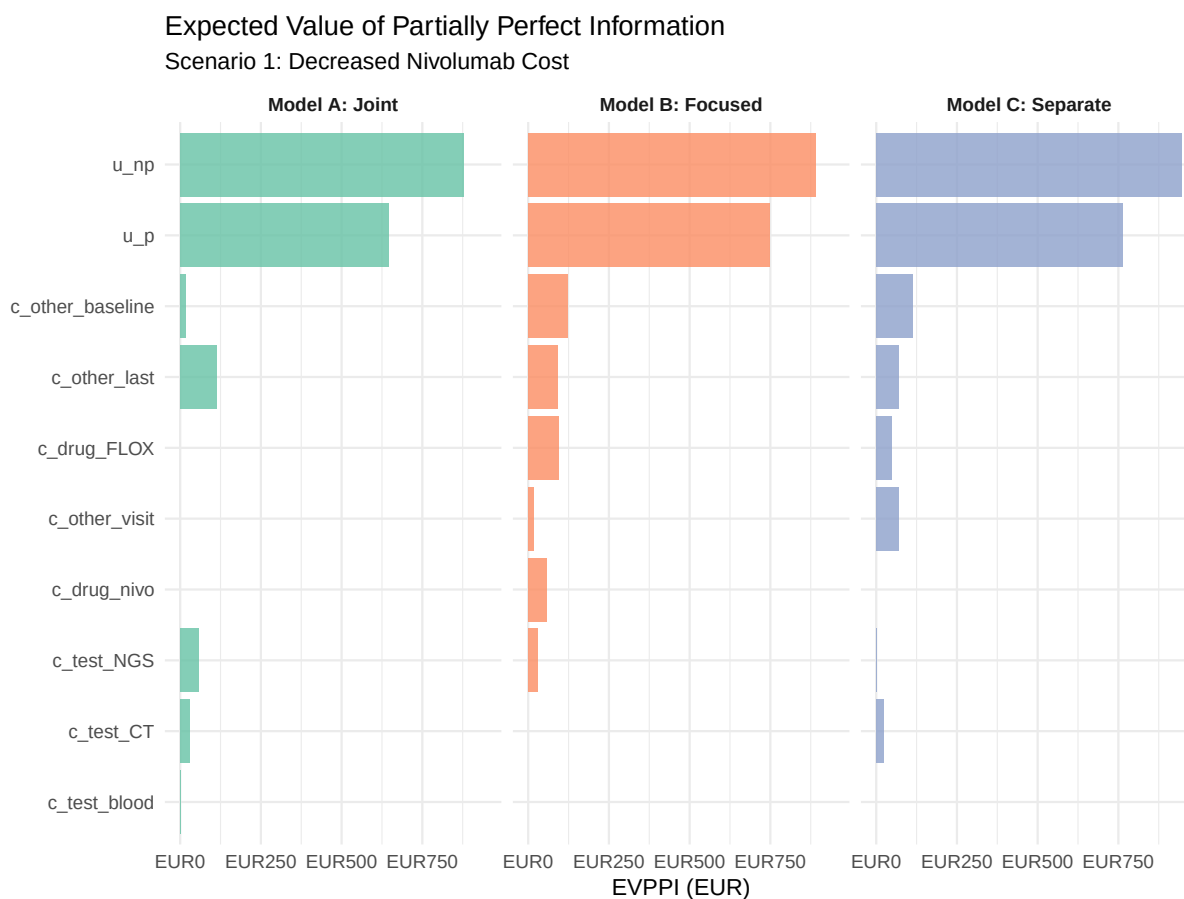


Figure 3: EVPPPI by parameter and model structure (Scenario 1)

6 Scenario 2: Higher WTP Threshold (EUR100,000)

Willingness-to-Pay Threshold: EUR1e+05

Nivolumab Cost: EUR13,923 (standard)

6.1 EVPI Comparison Across Model Structures

Table 7: Expected Value of Perfect Information (EVPI) by model structure - Scenario 2

	Model	EVPI
52	Model A: Joint	EUR10920.48
521	Model B: Focused	EUR11721.16
622	Model C: Separate	EUR10355.1

6.2 EVPPI Results by Model Structure

Table 8: EVPPI results by model structure - Scenario 2: WTP EUR100,000

Parameter	EVPPi (A)	% (A)	EVPPi (B)	% (B)	EVPPi (C)	% (C)
u_np	EUR1792.2	16.4%	EUR1906.43	16.3%	EUR1769.43	17.1%
u_p	EUR1443.3	13.2%	EUR1775.68	15.1%	EUR1811.9	17.5%
c_other_follow	EUR172.54	1.6%	EUR47.35	0.4%	EUR142.76	1.4%
c_test_CT	EUR139.69	1.3%	EUR36.34	0.3%	EUR137.86	1.3%
c_drug_FLOX	EUR59.49	0.5%	EUR0	0%	EUR144.05	1.4%
c_drug_nivo	EUR0	0%	EUR173.99	1.5%	EUR0	0%
c_other_last	EUR81.77	0.7%	EUR0	0%	EUR46.62	0.5%
c_other_visit	EUR0	0%	EUR0	0%	EUR98.84	1%
c_test_NGS	EUR0	0%	EUR90.67	0.8%	EUR0	0%
c_test_blood	EUR0	0%	EUR0	0%	EUR53.26	0.5%
c_other_baseline	EUR0	0%	EUR0	0%	EUR8.68	0.1%

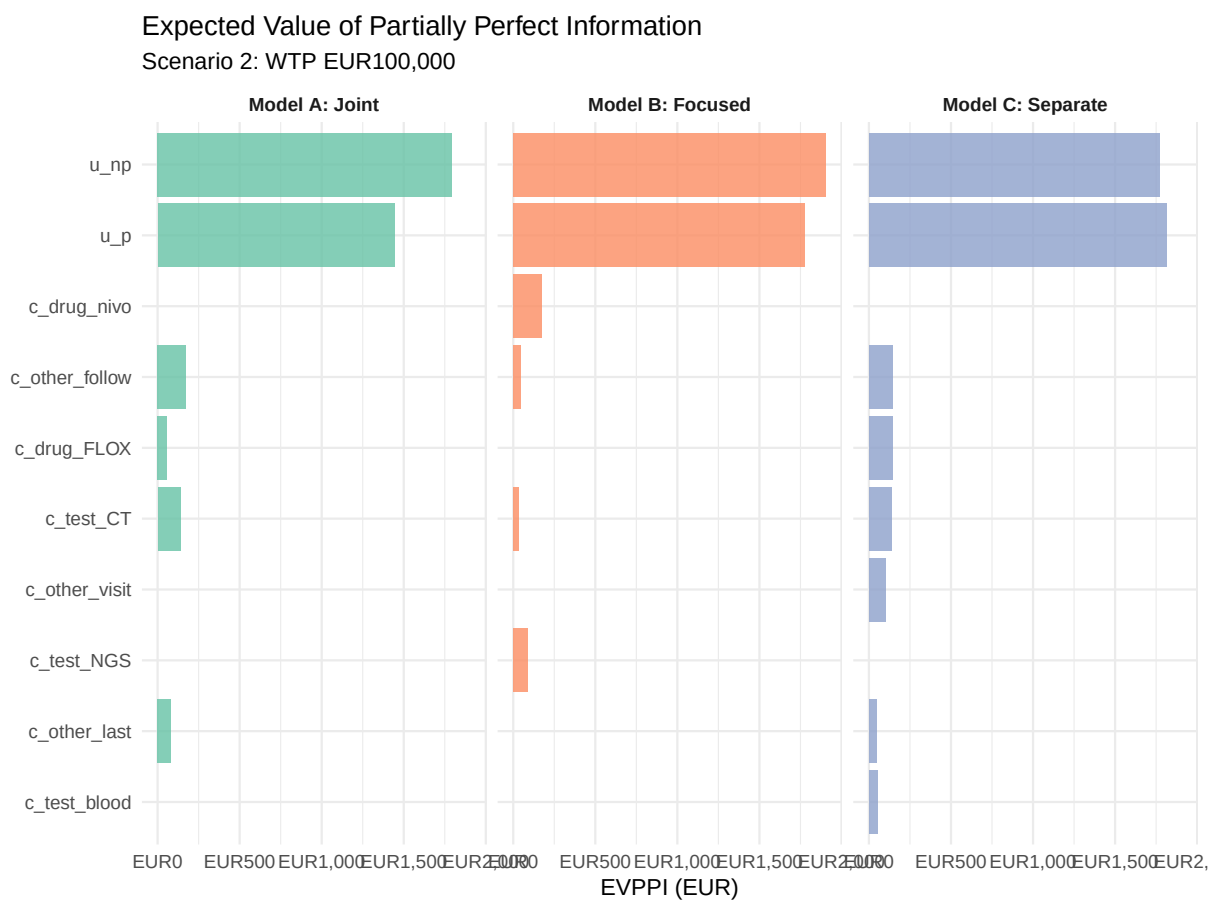


Figure 4: EVPPi by parameter and model structure (Scenario 2)

7 Scenario 3: Higher WTP with Decreased Nivolumab Cost

Willingness-to-Pay Threshold: EUR1e+05

Nivolumab Cost: EUR4,641 (80% discount)

7.1 EVPI Comparison Across Model Structures

Table 9: Expected Value of Perfect Information (EVPI) by model structure - Scenario 3

	Model	EVPI
53	Model A: Joint	EUR18084.2
531	Model B: Focused	EUR19218.52
532	Model C: Separate	EUR17658.69

7.2 EVPPI Results by Model Structure

Table 10: EVPPI results by model structure - Scenario 3: WTP EUR100,000 + Decreased Nivolumab

Parameter	EVPI (A)	% (A)	EVPI (B)	% (B)	EVPI (C)	% (C)
u_np	EUR2042.18	11.3%	EUR2367.58	12.3%	EUR2504.86	14.2%
u_p	EUR1712.51	9.5%	EUR1433.42	7.5%	EUR1500.91	8.5%
c_other_visit	EUR243.45	1.3%	EUR0	0%	EUR19.99	0.1%
c_drug_FLOX	EUR179.2	1%	EUR0	0%	EUR0	0%
c_other_baseline	EUR0	0%	EUR0	0%	EUR115.44	0.7%
c_test_blood	EUR0	0%	EUR0	0%	EUR63.33	0.4%
c_other_follow	EUR0	0%	EUR52.03	0.3%	EUR0	0%
c_drug_nivo	EUR22.43	0.1%	EUR0	0%	EUR27.27	0.2%
c_other_last	EUR45.9	0.3%	EUR0	0%	EUR0	0%
c_test_NGS	EUR0	0%	EUR29.51	0.2%	EUR12.03	0.1%
c_test_CT	EUR0	0%	EUR0	0%	EUR0	0%

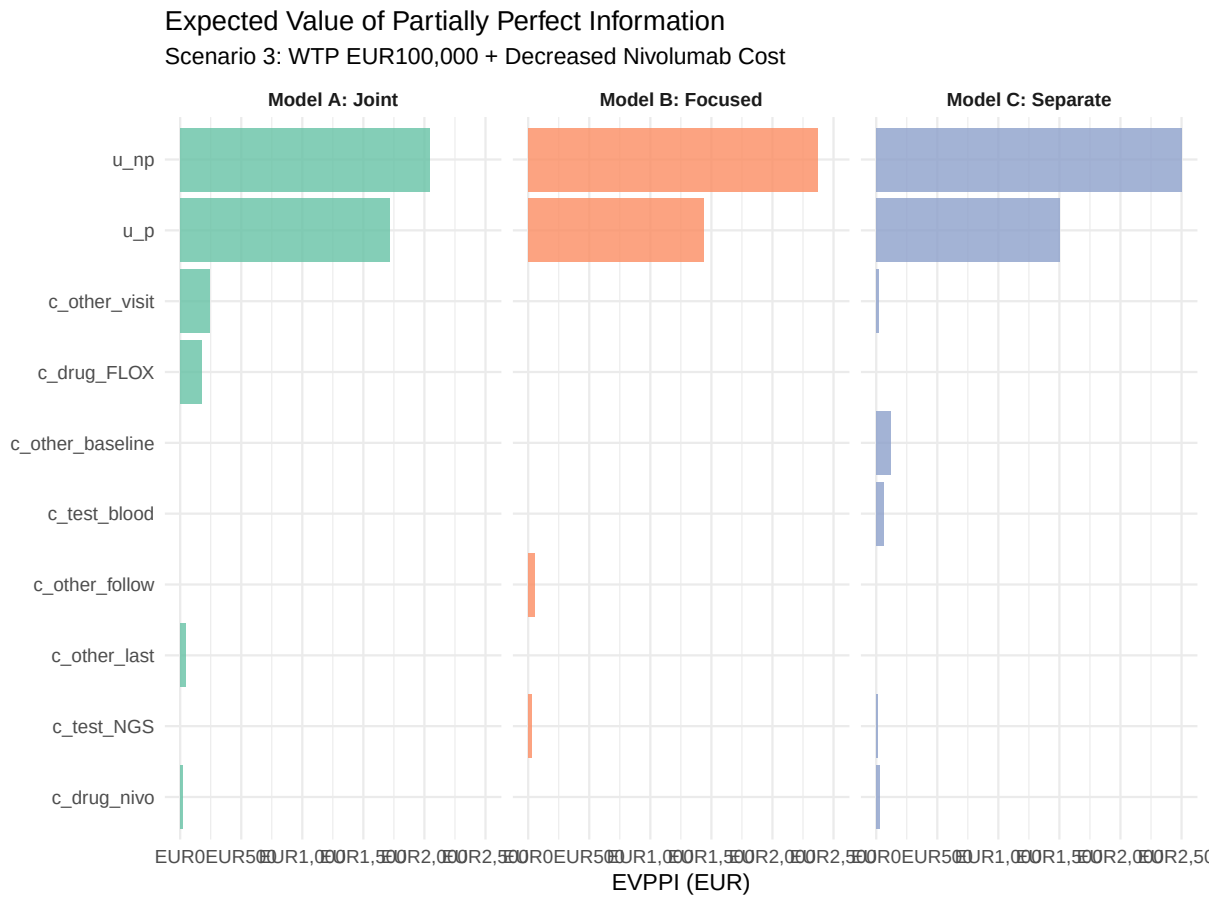


Figure 5: EVVPI by parameter and model structure (Scenario 3)

8 Scenario 4: Highest WTP Threshold (EUR150,000)

Willingness-to-Pay Threshold: EUR150,000

Nivolumab Cost: EUR13,923 (standard)

8.1 EVPI Comparison Across Model Structures

Table 11: Expected Value of Perfect Information (EVPI) by model structure - Scenario 4

Model		EVPI
54	Model A: Joint	EUR26196.17
541	Model B: Focused	EUR26609.53
542	Model C: Separate	EUR25651.06

8.2 EVPPI Results by Model Structure

Table 12: EVPPI results by model structure - Scenario 4: WTP EUR150,000

Parameter	EVPPI (A)	% (A)	EVPPI (B)	% (B)	EVPPI (C)	% (C)
u_np	EUR2850.76	10.9%	EUR2880.14	10.8%	EUR2605.4	10.2%
u_p	EUR2569.5	9.8%	EUR2193.62	8.2%	EUR2290.87	8.9%
c_drug_nivo	EUR357.15	1.4%	EUR292.74	1.1%	EUR606.72	2.4%
c_other_follow	EUR130.39	0.5%	EUR49.95	0.2%	EUR312.42	1.2%
c_other_visit	EUR220.84	0.8%	EUR236.05	0.9%	EUR0	0%
c_test_NGS	EUR26.79	0.1%	EUR243.16	0.9%	EUR0	0%
c_test_blood	EUR81.63	0.3%	EUR0	0%	EUR18.28	0.1%
c_test_CT	EUR88.21	0.3%	EUR0	0%	EUR0	0%
c_drug_FLOX	EUR0	0%	EUR0.66	0%	EUR0	0%
c_other_last	EUR0	0%	EUR0	0%	EUR0	0%
c_other_baseline	EUR0	0%	EUR0	0%	EUR0	0%

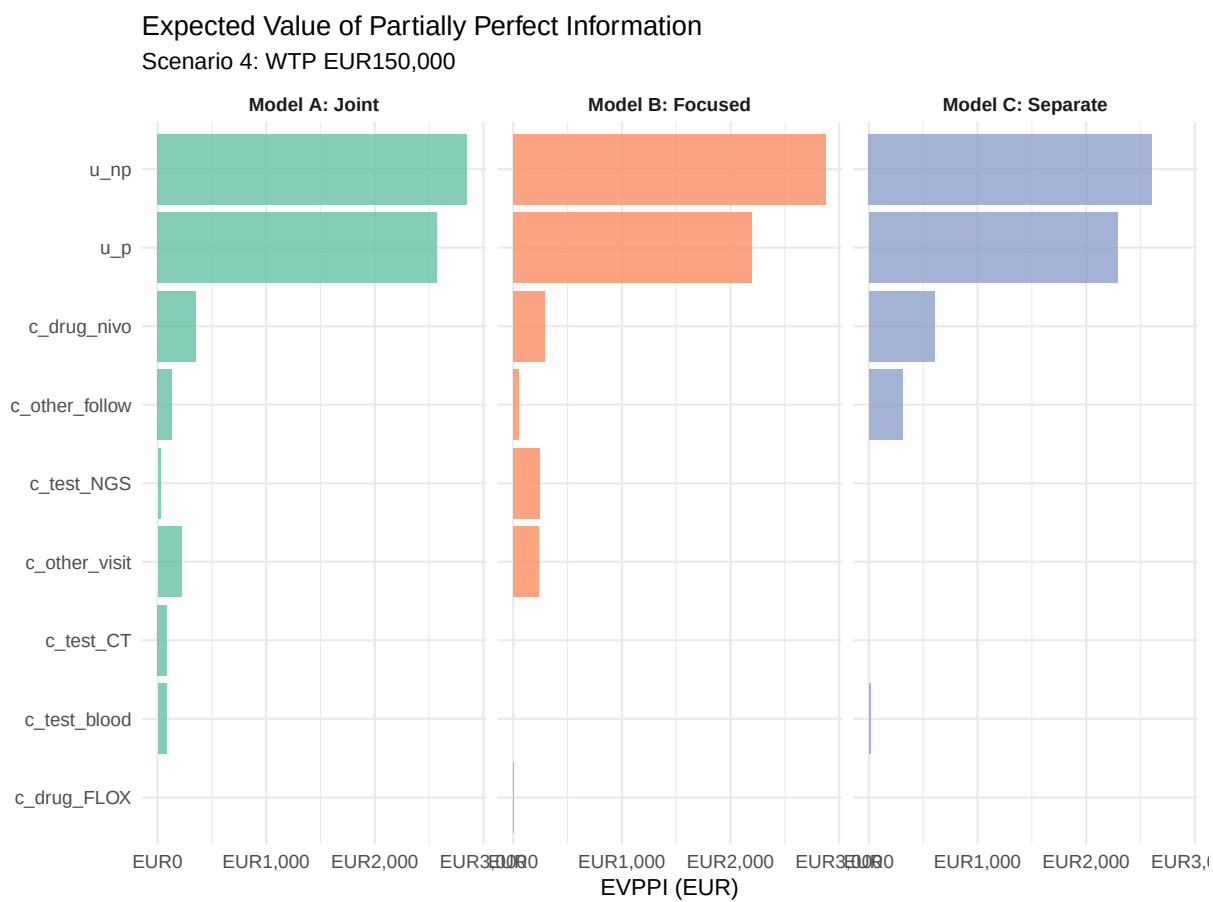


Figure 6: EVPPI by parameter and model structure (Scenario 4)

9 Scenario 5: Highest WTP with Decreased Nivolumab Cost

Willingness-to-Pay Threshold: EUR150,000

Nivolumab Cost: EUR4,641 (80% discount)

9.1 EVPI Comparison Across Model Structures

Table 13: Expected Value of Perfect Information (EVPI) by model structure - Scenario 5

	Model	EVPI
55	Model A: Joint	EUR25078.7
551	Model B: Focused	EUR28398.56
552	Model C: Separate	EUR24190.09

9.2 EVPPI Results by Model Structure

Table 14: EVPPI results by model structure - Scenario 5: WTP EUR150,000 + Decreased Nivolumab

Parameter	EVPI (A)	% (A)	EVPI (B)	% (B)	EVPI (C)	% (C)
u_np	EUR3231.38	12.9%	EUR3519.42	12.4%	EUR3716.18	15.4%
u_p	EUR2694.5	10.7%	EUR2386.86	8.4%	EUR1914.54	7.9%
c_drug_FLOX	EUR241.08	1%	EUR117.76	0.4%	EUR298.93	1.2%
c_other_last	EUR229.42	0.9%	EUR0	0%	EUR254.72	1.1%
c_other_visit	EUR240.34	1%	EUR189.38	0.7%	EUR0	0%
c_test_NGS	EUR0	0%	EUR186.32	0.7%	EUR154.78	0.6%
c_other_follow	EUR146.54	0.6%	EUR47.9	0.2%	EUR94.64	0.4%
c_test_CT	EUR168.06	0.7%	EUR83.96	0.3%	EUR0	0%
c_other_baseline	EUR178.12	0.7%	EUR67.48	0.2%	EUR0	0%
c_drug_nivo	EUR114.14	0.5%	EUR4.53	0%	EUR16.52	0.1%
c_test_blood	EUR0	0%	EUR69.26	0.2%	EUR0	0%

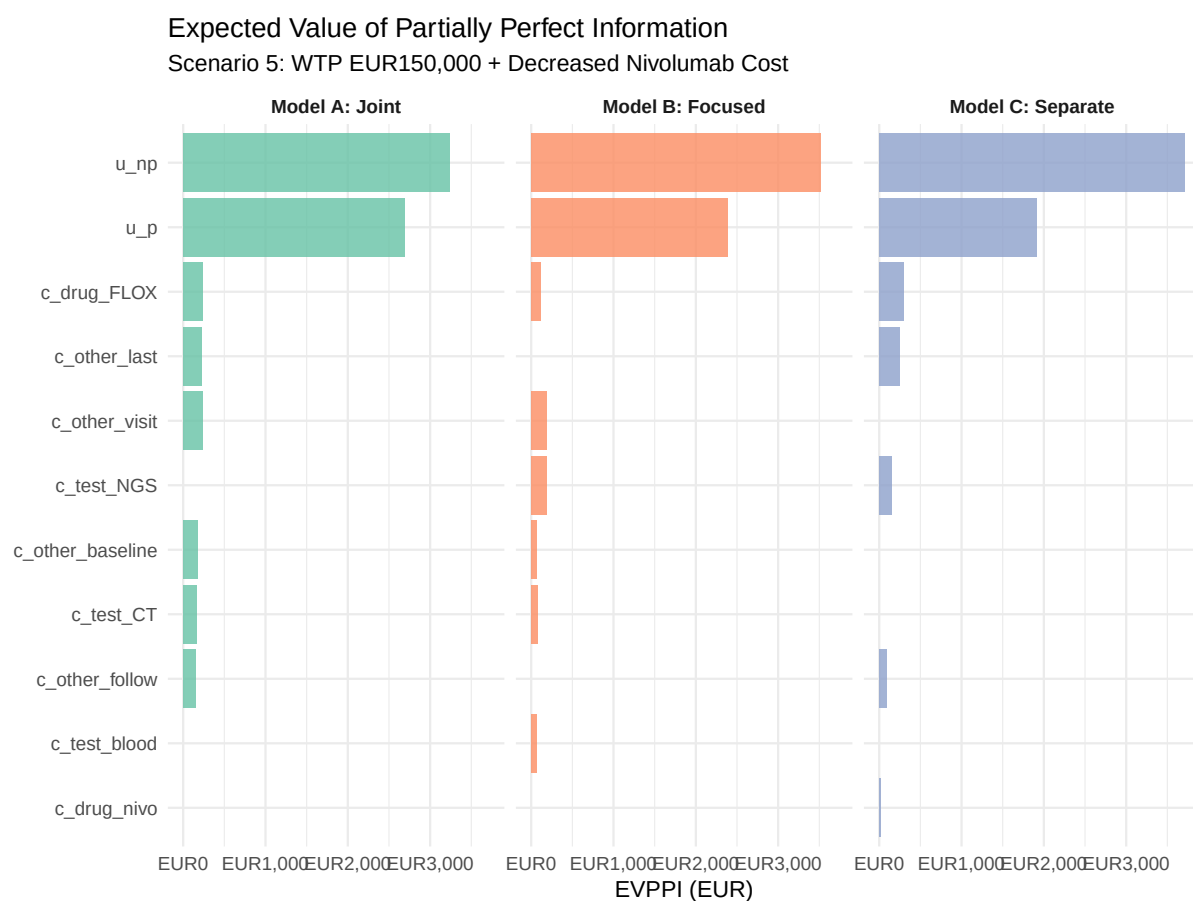


Figure 7: EVVPI by parameter and model structure (Scenario 5)

10 Comparative Analysis Across Scenarios and Models

10.1 EVPI Comparison



Figure 8: Expected Value of Perfect Information across scenarios and model structures

10.2 EVPPI Parameter Heatmap by Model Structure

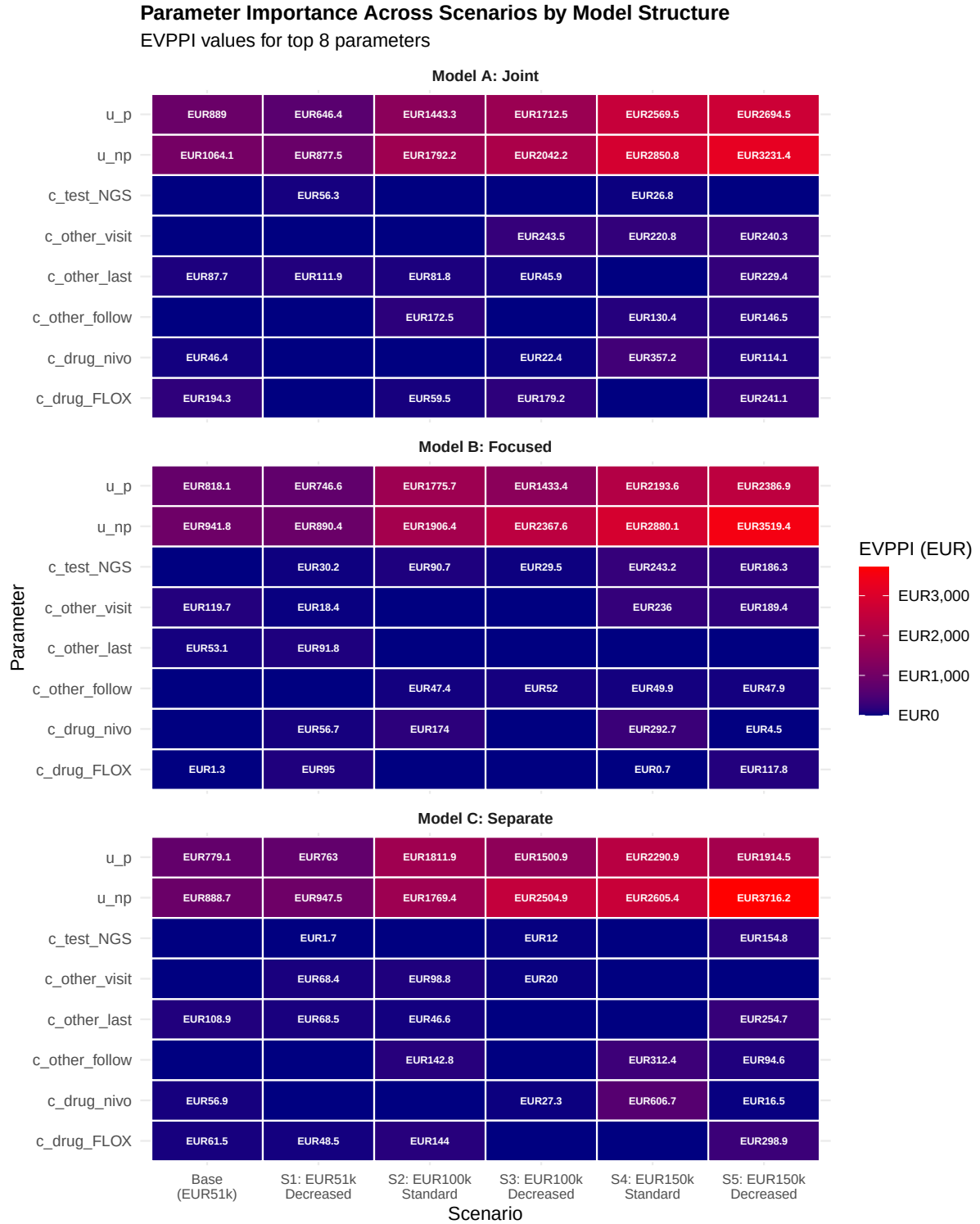


Figure 9: Parameter importance across scenarios by model structure (top 8 parameters)

10.3 Summary Comparison Table

Table 15: Summary of EVPI and top EVPPI by model structure and scenario

Model	Scenario	WTP	EVPI	Top Parameter	Max EVPPI	% of EVPI
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Model A: Joint	Base Case	EUR 51,000	EUR1339.56	u_np	EUR1064.09	79.4
Model A: Joint	Decreased Nivolumab Cost	EUR 51,000	EUR8696.69	u_np	EUR877.52	10.1
Model A: Joint	WTP €100,000	EUR100,000	EUR10920.48	u_np	EUR1792.2	16.4
Model A: Joint	WTP €100,000 + Decreased Nivolumab	EUR100,000	EUR18084.2	u_np	EUR2042.18	11.3
Model A: Joint	WTP €150,000	EUR150,000	EUR26196.17	u_np	EUR2850.76	10.9
Model A: Joint	WTP €150,000 + Decreased Nivolumab	EUR150,000	EUR25078.7	u_np	EUR3231.38	12.9
Model B: Focused	Base Case	EUR 51,000	EUR1438.25	u_np	EUR941.75	65.5
Model B: Focused	Decreased Nivolumab Cost	EUR 51,000	EUR8946	u_np	EUR890.43	10.0
Model B: Focused	WTP €100,000	EUR100,000	EUR11721.16	u_np	EUR1906.43	16.3
Model B: Focused	WTP €100,000 + Decreased Nivolumab	EUR100,000	EUR19218.52	u_np	EUR2367.58	12.3
Model B: Focused	WTP €150,000	EUR150,000	EUR26609.53	u_np	EUR2880.14	10.8
Model B: Focused	WTP €150,000 + Decreased Nivolumab	EUR150,000	EUR28398.56	u_np	EUR3519.42	12.4
Model C: Separate	Base Case	EUR 51,000	EUR922.7	u_np	EUR888.69	96.3
Model C: Separate	Decreased Nivolumab Cost	EUR 51,000	EUR8537.63	u_np	EUR947.52	11.1
Model C: Separate	WTP €100,000	EUR100,000	EUR10355.1	u_p	EUR1811.9	17.5
Model C: Separate	WTP €100,000 + Decreased Nivolumab	EUR100,000	EUR17658.69	u_np	EUR2504.86	14.2
Model C: Separate	WTP €150,000	EUR150,000	EUR25651.06	u_np	EUR2605.4	10.2
Model C: Separate	WTP €150,000 + Decreased Nivolumab	EUR150,000	EUR24190.09	u_np	EUR3716.18	15.4

10.4 Key Findings

10.4.1 Model Structure Robustness

Parameter importance is consistent across model structures. The parameter ‘u_np’ is the most important driver of value of information across all three model specifications. This suggests that conclusions about research priorities are robust to the choice of model structure.

10.4.2 Scenario with Greatest Decision Uncertainty

- **Model A: Joint:** WTP €150,000 (EVPI = EUR26196.17)
- **Model B: Focused:** WTP €150,000 + Decreased Nivolumab (EVPI = EUR28398.56)
- **Model C: Separate:** WTP €150,000 (EVPI = EUR25651.06)

10.4.3 Most Important Parameter Across All Scenarios

u_np (Total EVPPI = EUR36795.99)

10.4.4 General Observations

Impact of WTP Threshold: Higher WTP thresholds generally increase decision uncertainty (EVPI), creating greater value for additional research to reduce parameter uncertainty.

Impact of Nivolumab Cost: Decreased nivolumab costs shift the cost-effectiveness landscape and may change which parameters drive decision uncertainty.

Impact of Model Structure: Comparing EVPI and EVPPI across model structures helps identify whether value of information conclusions are robust to structural uncertainty in the survival model specification.

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